

Examiner
only

7. Magnesium chloride is a solid with a high melting point. It conducts electricity only when molten or dissolved.

Describe the electronic changes that take place in the formation of magnesium chloride and explain its properties in terms of the bonding and structure. You may use diagrams as part of your answer. [6 QER]

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END OF PAPER

6



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		





24

THE PERIODIC TABLE

Group 1 2 3 4 5 6 7 0

1H Hydrogen 1																		24	
7Li Lithium 3	9Be Beryllium 4																	20Ne Neon 10	4He Helium 2
23Na Sodium 11	24Mg Magnesium 12																	19F Fluorine 9	
39K Potassium 19	40Ca Calcium 20	45Sc Scandium 21	48Ti Titanium 22	51V Vanadium 23	52Cr Chromium 24	55Mn Manganese 25	56Fe Iron 26	59Co Cobalt 27	59Ni Nickel 28	63.5Cu Copper 29	65Zn Zinc 30	70Ga Gallium 31	73Ge Germanium 32	75As Arsenic 33	79Se Selenium 34	80Br Bromine 35	35.5Cl Chlorine 17	40Ar Argon 18	
86Rb Rubidium 37	88Sr Strontium 38	89Y Yttrium 39	91Zr Zirconium 40	93Nb Niobium 41	96Mo Molybdenum 42	99Tc Technetium 43	101Ru Ruthenium 44	103Rh Rhodium 45	106Pd Palladium 46	108Ag Silver 47	112Cd Cadmium 48	115In Indium 49	119Sn Tin 50	122Sb Antimony 51	128Te Tellurium 52	127I Iodine 53		131Xe Xenon 54	
133Cs Caesium 55	137Ba Barium 56	139La Lanthanum 57	179Hf Hafnium 72	181Ta Tantalum 73	184W Tungsten 74	186Re Rhenium 75	190Os Osmium 76	192Ir Iridium 77	195Pt Platinum 78	197Au Gold 79	201Hg Mercury 80	204Tl Thallium 81	207Pb Lead 82	209Bi Bismuth 83	210Po Polonium 84	210At Astatine 85		222Rn Radon 86	
223Fr Francium 87	226Ra Radium 88	227Ac Actinium 89																	

Key

A_r

Symbol

Name

Z

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
Other Names		0



GCSE
3430UE0-1
SCIENCE (Double Award)



Unit 5 – CHEMISTRY 2
HIGHER TIER

THURSDAY, 16 MAY 2019 – MORNING
1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	15	
2.	7	
3.	6	
4.	15	
5.	11	
6.	6	
Total	60	

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

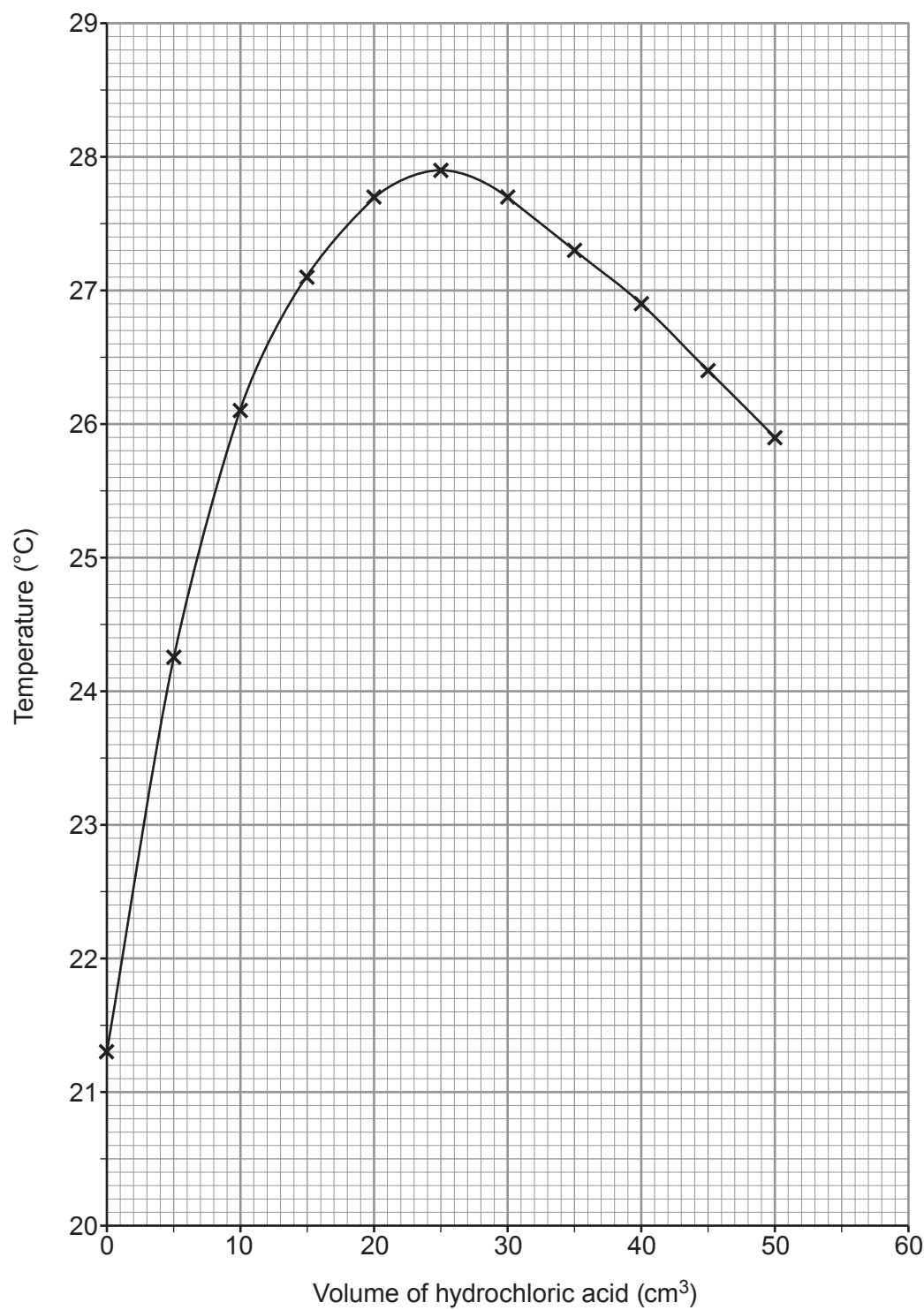
INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.
The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.



MAY193430UE0101

7. (a) The graph shows how the temperature changes when dilute hydrochloric acid is added gradually to sodium hydroxide solution.



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The pH of the solution also changes as the hydrochloric acid is added to the sodium hydroxide.

Explain the temperature changes shown on the graph and relate these to the change in pH during the reaction. You do **not** need to include equations in your answer. [6 QER]

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(b) The reaction between hydrochloric acid and sodium hydroxide produces sodium chloride and water.

Give the **ionic** equation to show how water is formed during this reaction. [1]

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END OF PAPER

7



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potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



Group

Group	3	4	5	6	7	0
1						
2						

[illegible]

Key

A_r	relative atomic mass
Symbol	
Name	
Z	atomic number

Surname	Centre Number	Candidate Number
First name(s)		0



GCSE
3430UE0-1



MONDAY, 22 MAY 2023 – MORNING

SCIENCE (Double Award)
Unit 5 – CHEMISTRY 2
HIGHER TIER
1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	15	
2.	7	
3.	7	
4.	9	
5.	10	
6.	6	
7.	6	
Total	60	

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

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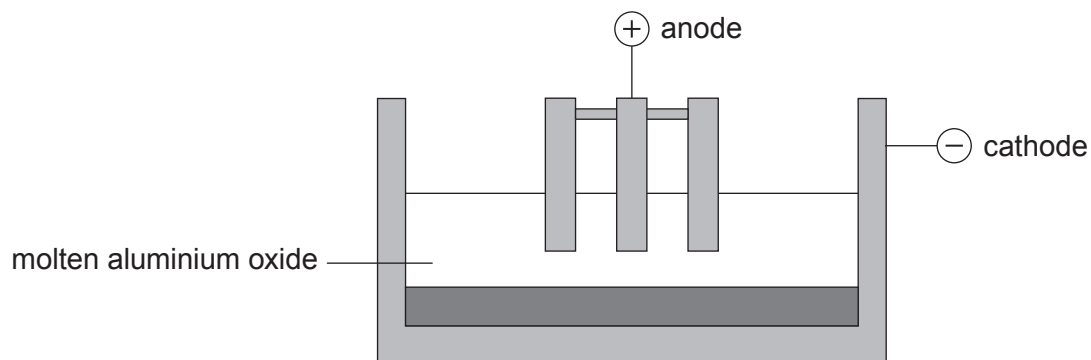
Question 7 is a quality of extended response (QER) question where your writing skills will be assessed.

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JUN233430UE0101

7. Aluminium is extracted from its ore by electrolysis. The diagram shows a cross-section of the cell used for the extraction.



The overall equation for the extraction taking place inside the cell is given below.



Examiner
only

Explain the movement of Al^{3+} and O^{2-} ions inside the cell and how the reactions at each electrode demonstrate oxidation and reduction. Include electrode equations in your answer.

[6 QER]

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silver	Ag^+		
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zinc	Zn^{2+}		



THE PERIODIC TABLE

Group 1 2 3 4 5 6 7 0



28

1

H

Hydrogen

1

7 Li Lithium 3		9 Be Beryllium 4		10 Ne Neon 10																															
23 Na Sodium 11		24 Mg Magnesium 12		35.5 Cl Chlorine 17																															
39 K Potassium 19		40 Ca Calcium 20		45 Sc Scandium 21		48 Ti Titanium 22		51 V Vanadium 23		52 Cr Chromium 24		55 Mn Manganese 25		56 Fe Iron 26		59 Co Cobalt 27		59 Ni Nickel 28		63.5 Cu Copper 29		65 Zn Zinc 30		70 Ga Gallium 31		73 Ge Germanium 32		75 As Arsenic 33		79 Se Selenium 34		80 Br Bromine 35		84 Kr Krypton 36	
86 Rb Rubidium 37		88 Sr Strontium 38		89 Y Yttrium 39		91 Zr Zirconium 40		93 Nb Niobium 41		96 Mo Molybdenum 42		99 Tc Technetium 43		101 Ru Ruthenium 44		103 Rh Rhodium 45		106 Pd Palladium 46		108 Ag Silver 47		112 Cd Cadmium 48		115 In Indium 49		119 Sn Tin 50		122 Sb Antimony 51		128 Te Tellurium 52		127 I Iodine 53		131 Xe Xenon 54	
133 Cs Caesium 55		137 Ba Barium 56		139 La Lanthanum 57		179 Hf Hafnium 72		181 Ta Tantalum 73		184 W Tungsten 74		186 Re Rhenium 75		190 Os Osmium 76		192 Ir Iridium 77		195 Pt Platinum 78		197 Au Gold 79		201 Hg Mercury 80		204 Tl Thallium 81		207 Pb Lead 82		209 Bi Bismuth 83		210 Po Polonium 84		210 At Astatine 85		222 Rn Radon 86	
223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89		Key																													

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number